

Lab_05

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Data Exploration and Visualization in R

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Notes: **stuff in bold** is added by using two asterisks on each side of text *something italic* is added by using one asterisk on each side of text. Indents are added into format by using the greater than sign

Indent Text

Sections 1-5 Below were done together as a class

Example of how to format questions and answers > Q1: Make a ggplot plot

Answer Q1: falsdfhjkasjd

```
# Example on how to make a vector and add one to every number in vector x  
x <- 1:5  
x+1
```

```
[1] 2 3 4 5 6
```

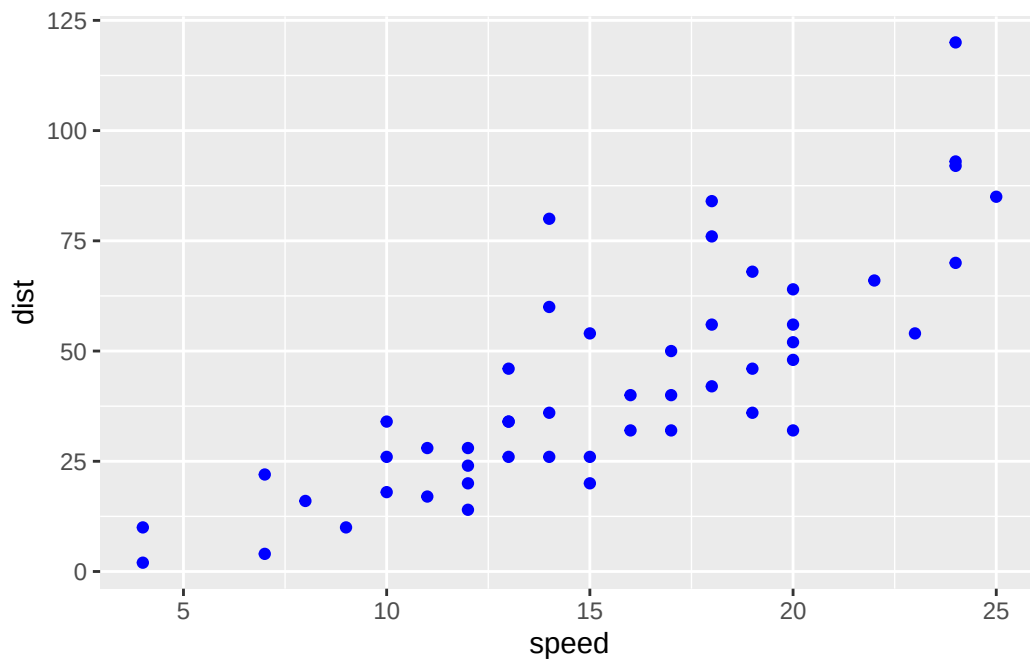
Let's Make a graph library(ggplot2) only has to be called out once. (cars) refers to the dataset were working with, aes() helps label things such as x and y axis, and geom_point() is the function used to plot the points on the graph itself.

```
# Call out ggplot2 so that we can load this package onto R

library(ggplot2)

# Use the cars data frame and functions like aes() and geom_point() to
# successfully label and generate a scatter plot of our data set.

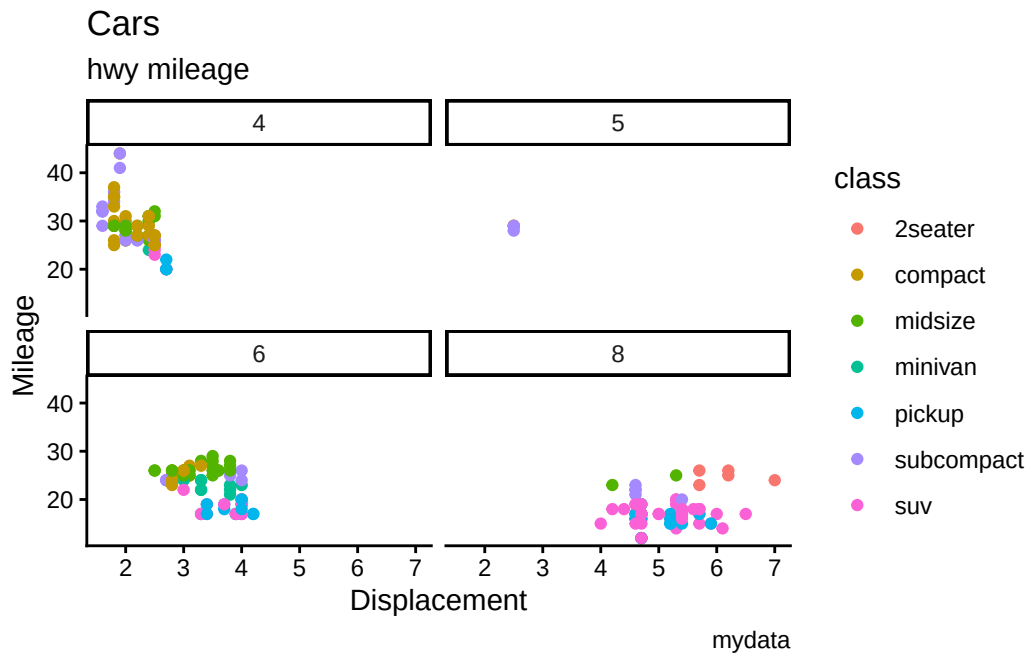
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point(col="BLUE")
```



Example of more modifications and changes made to the appearance of the graph

```
# Practice adding modifications and changes to the appearance of our graphs.
# Modify theme for clarity, add labels to make chart easier to understand.

ggplot(mpg) +
  aes(x=displ, y=hwy, col=class) +
  geom_point()+
  theme_classic()+
  labs(title="Cars", subtitle="hwy mileage", caption="mydata", x="Displacement", y="Mileage")
  facet_wrap(~cyl)
```

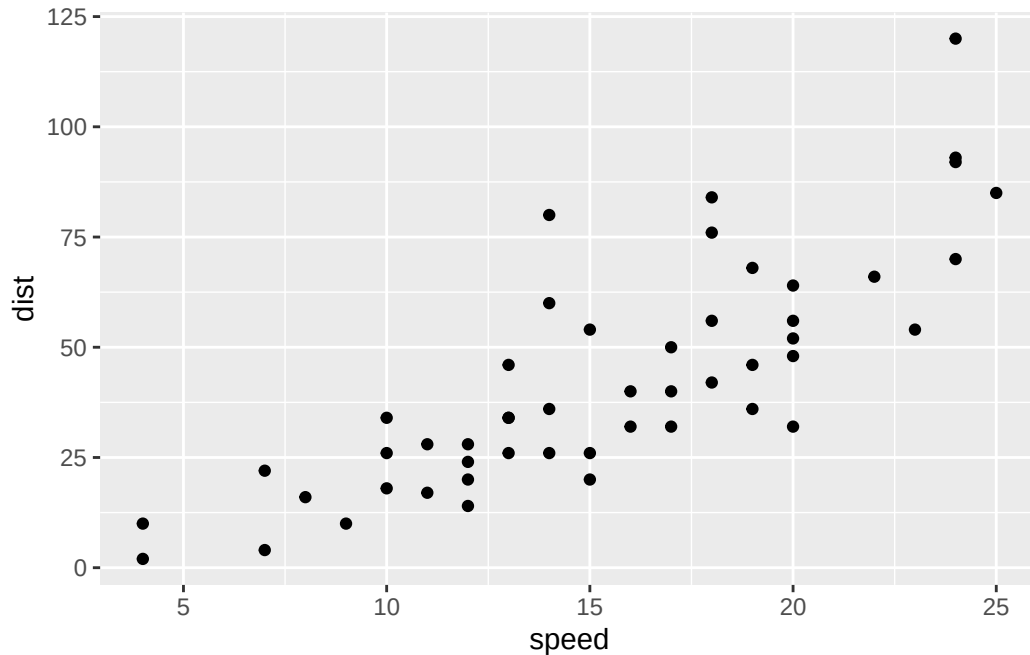


Start of Section 6: Creating Scatter Plots

We begin by mapping speed and distance variables taken from the cars dataset as shown below.

```
# Map the data set "cars" and display it in a scatter plot

ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point()
```



Question 1: Which geometric layer should be used to create scatter plots in ggplot2?

Answer Q1: In ggplot 2, `geom_point()` is the geometric layer used to create scatter plots like the one shown above. The full r code that accounts for this plot is shown above.

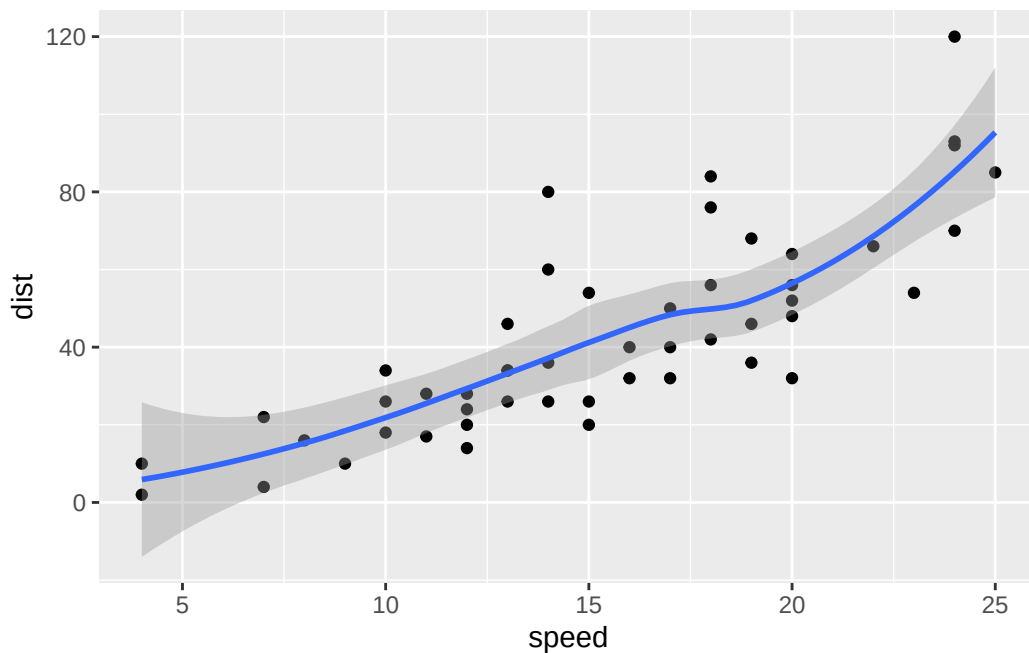
Question 2: In your own RStudio can you add a trend line layer to help show the relationship between the plot variables with the `geom_smooth()` function?

Answer Q2 Yes, a trend line layer can be added to the ggplot by introducing it into the code as `geom_smooth()`. This will generate a thin line that follows the trend of the plots which allows us to better visualize the positive association/relationship between speed and distance. The code for the addition of this trendline is shown below.

```
# Add the geom_smooth() function to add a trendline
```

```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point() +  
  geom_smooth()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



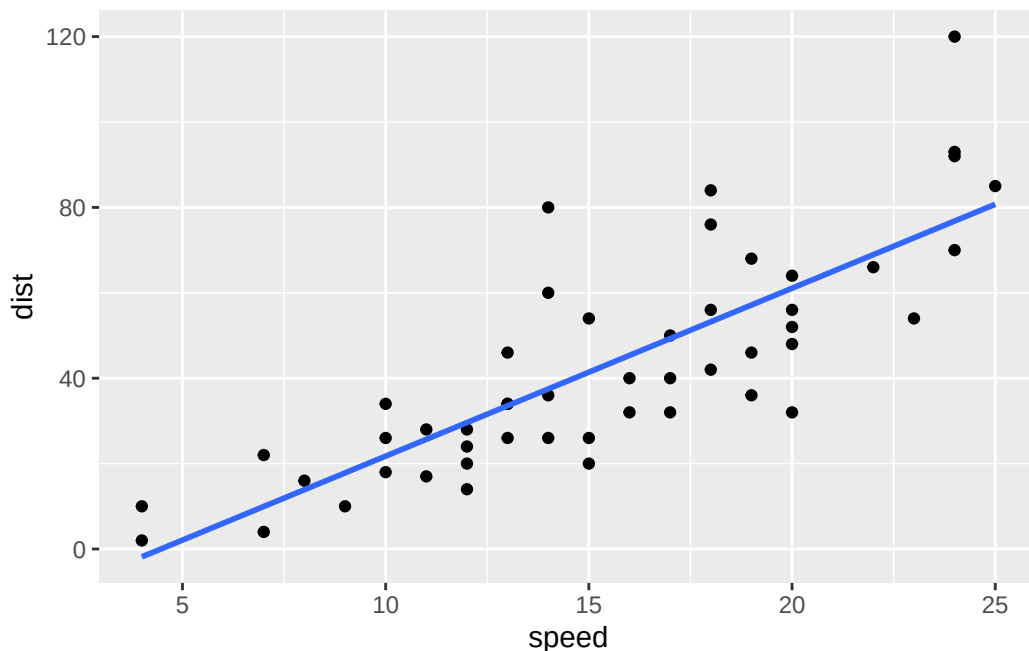
Question 3: Argue with `geom_smooth()` to add a straight line from a linear model without the shaded standard error region.

Answer Q3: In order to add a straight line without the shaded standard error region we must incorporate a new function to our `geom_smooth()` code. By adding `method="lm"` into our code we make the trend line linear and by adding `se=FALSE` we remove the standard error region from the plotted trend line. Together, these codes are added into `geom_smooth(method="lm", se=FALSE)` to generate the straight linear model without the shaded standard error region shown below.

```
# Add modifications in the geom_smooth() function to change the appearance of the trendline.
# Changing method to lm makes the line of best fit linear and making se=FALSE
# allows us to remove the grey standard error of the trendline.
```

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE)
```

```
`geom_smooth()` using formula = 'y ~ x'
```



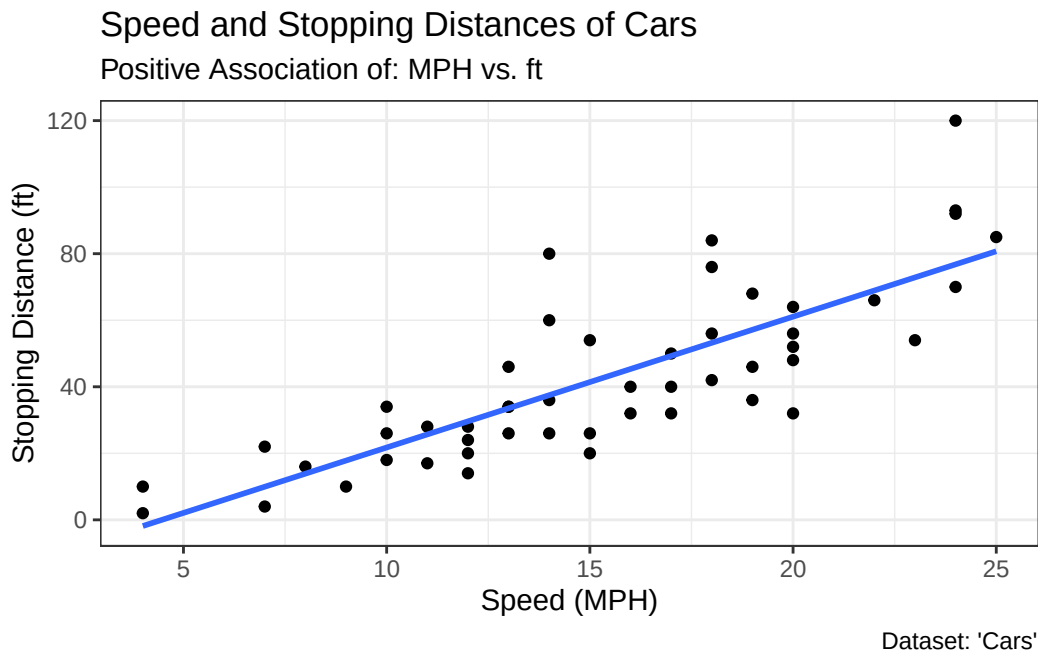
Question 4: Can you finish this plot by adding various label annotations with the `labs()` function and the plot look to a more conservative “black & white” theme by adding the `theme_bw()` function?

Answer Q4: By using the `labs()` function labels such as the graph title and x and y axis titles can be incorporated onto our graph. Adding these labels makes our graph easier to follow, interpret, and analyze. By adding the `theme_bw()` function, our graph is changed to a black and white theme making our graph “cleaner” and more conservative by avoiding background noise in colors. The finished plot and its corresponding r code are shown below.

```
# Use the labs() function to annotate the display of our plot. Adding labels.

ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE) +
  labs(title="Speed and Stopping Distances of Cars",
       x="Speed (MPH)",
       y="Stopping Distance (ft)",
       subtitle = "Positive Association of: MPH vs. ft",
       caption="Dataset: 'Cars'") +
  theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Example Data Set of Gene Table

```
# This set is included to access a new dataset to help answer the
# following questions.

url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

Question 5: Use the `nrow()` function to find out how many genes are in the data set. What is your answer?

Answer Q5: There are a total of 5,196 genes in the data set above. The r code used to find this calculation is shown below.

```
nrow(genes)
```

```
[1] 5196
```

Question 6: Use the `colnames()` function and the `ncol()` function on the genes data frame to find out what the column names are (we will find these later) and how many columns there are. How many columns did you find?

Answer Q6: There are a total of 4 columns in the genes data set. The names of these 4 columns are: “Gene”, “Condition 1”, “Condition 2”, and “State”. The r code used to find this information is shown below.

```
#Practice manipulating functions to display only the information we need from  
# our data frame. In this case we only called out the column names and not the  
#components of each column as these do not matter at the moment.
```

```
colnames(genes)
```

```
[1] "Gene"          "Condition1" "Condition2" "State"
```

```
ncol(genes)
```

```
[1] 4
```

Question 7: Use the `table()` function on the “State” column of this data.frame to find out how many “up” regulated genes there are. What is your answer?

Answer Q7: There are a total of 127 up-regulated genes in this data.frame. The r code used to find this calculation is: `table(genes$State)`. We use `genes` to show that this we are searching in the genes data set, then we use `$State` to call out the State column only.

```
# Use $ to call out a single section of the data set, in this case only  
# factor information from the State Column
```

```
table(genes$State)
```


down	unchanging	up
72	4997	127

Question 8: Using your values above and 2 significant figures. What *percent* of total genes is up-regulated in this dataset?

Answer Q8: The percent of the of total genes that are up-regulated in this data set is: 2.44%. We take the total number of up-regulated genes then divide it by the total number of genes and then multiply this number to get the percentage: $(127/5,196) \times 100$. The r code for this calculation is shown below.

```
# Use round(...,2) to round the valyes in our argument to 2 significant
# figures. Our argument is the calculation shown below.
```

```
round(table(genes$State)/nrow(genes) * 100, 2 ) [3]
```

```
up
2.44
```

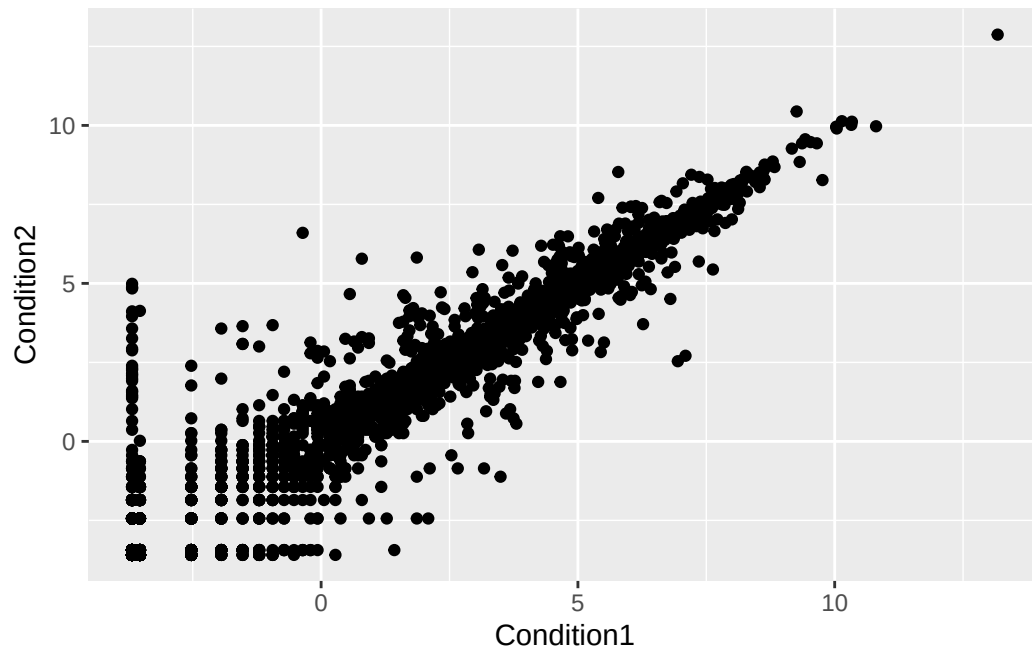
First Basic Scatter plot of this Dataset

Question 9: Complete the code below to produce the following plot: `ggplot() + aes(x=Condition1, y=)` + _____

Answer Q9: The completed version of the code is shown in the r code bewlow. In this plot we are comparing Condition 1 and Condition 2, resulting in a positive association.

```
# More practice on generating scatter plots
```

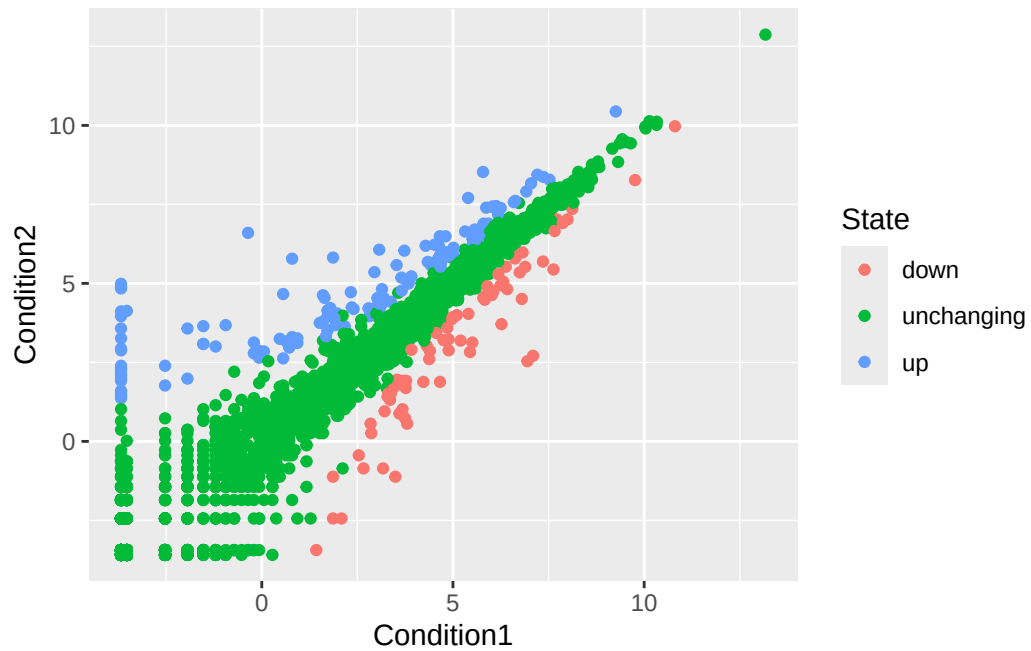
```
ggplot(genes) +
  aes(x=Condition1, y=Condition2) +
  geom_point()
```



Map of State Column to point color

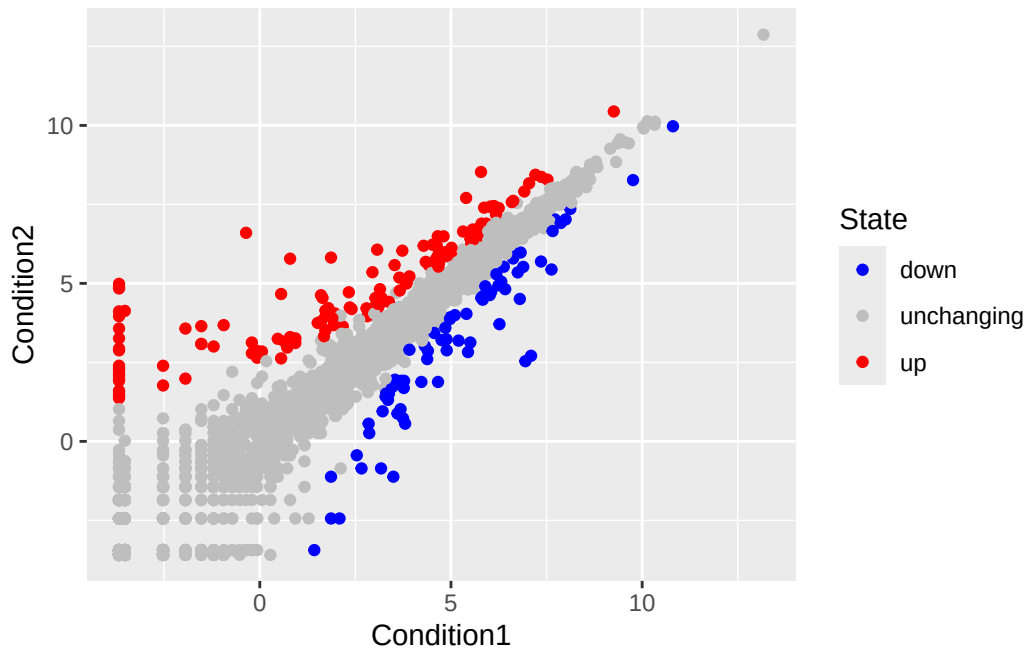
```
# Make this plot a vector to make it easier to access when making
# modifications in the future.  Add a color scale based on the State Column

p <- ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point()
p
```



Explicitly Specified Color Scale

```
# Use scale_color to adjust the color of our graph and make its visuals  
# centered on the most important results (Down and Up regulated genes)  
  
p + scale_colour_manual( values=c("blue","gray","red") )
```



Question 10: Nice, now add some plot annotations to the p object with the labs() function so your plot looks like the following: *We were shown a properly labeled axis and graph title*

Answer Q10: To add the Graph title and x and y axis title we much add the following function to our previous specified color scale code: + labs(title="Gene Expression Changes Upon Drug Treatment", x="Control (no drug) ", y="Drug Treatment")

```
# Practice adding labels to make our graph match up with the one
# shown in our manual. Use the names provided for every label

p + scale_colour_manual(values=c("blue","gray","red")) +
  labs(title="Gene Expression Changes Upon Drug Treatment",
       x="Control (no drug) ",
       y="Drug Treatment")
```

Gene Expression Changes Upon Drug Treatment

